

Urban Fledgelings Analysis

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PUBLISHED
April 24, 2026

RESEARCH QUESTION
How do environmental stresses and resource availability influence reproductive success?

1 prepare the session, import and wrangle the data [🔗](#)

2 explore the data

2.1 numerically

Table 1: Descriptive statistics of data in data dataframe.

	sample_id	number_of_fledglings	Insect.Abundance_g.m2	environmental_noise_dB
nbr.val	39.000	39.000	39.000	39.000
nbr.null	0.000	0.000	0.000	0.000
nbr.na	0.000	0.000	0.000	0.000
min	1.000	3.000	6.000	45.000
max	40.000	7.000	16.000	88.000
range	39.000	4.000	10.000	43.000
sum	797.000	182.000	406.000	2286.000
median	20.000	5.000	10.000	59.000
mean	20.436	4.667	10.410	58.615
SE.mean	1.895	0.209	0.471	1.276
CI.mean.0.95	3.837	0.423	0.954	2.583
var	140.094	1.702	8.669	63.506
std.dev	11.836	1.305	2.944	7.969
coef.var	0.579	0.280	0.283	0.136
skewness	0.016	0.267	0.118	0.954
skew.2SE	0.021	0.353	0.156	1.261
kurtosis	-1.331	-1.121	-1.219	2.634
kurt.2SE	-0.898	-0.756	-0.822	1.777
normtest.W	0.952	0.899	0.950	0.921
normtest.p	0.096	0.002	0.079	0.009

In [Table 1](#) we see that the P-value of the Shapiro-Wilk test for the outcome variable is < 0.05, indicating that the outcome variable isn't normally distributed. Furthermore, the outcome is count data, so we use a poisson GLM to model the data.

2.2 graphically

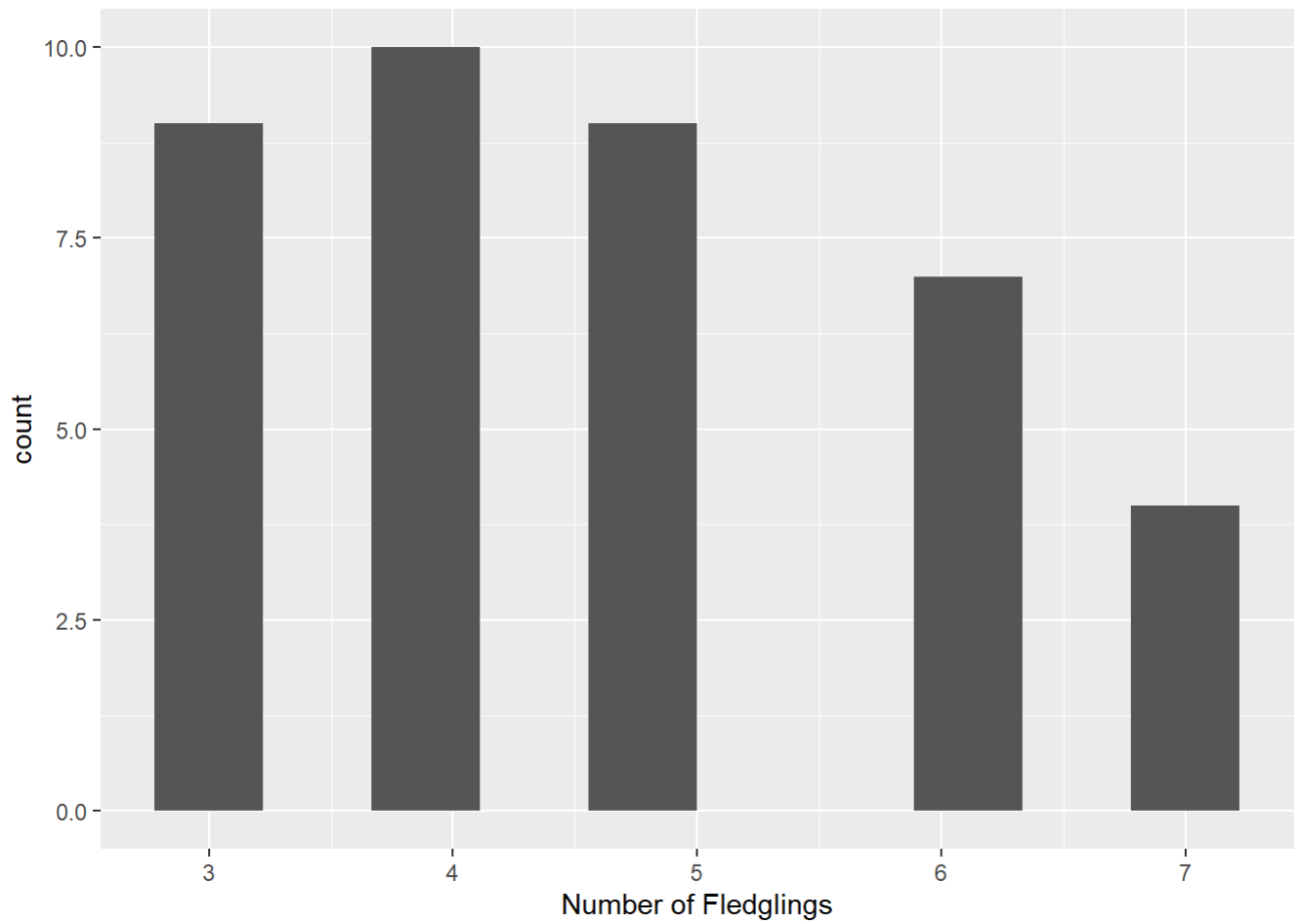


Figure 1: Distribution of number of fledglings observed.

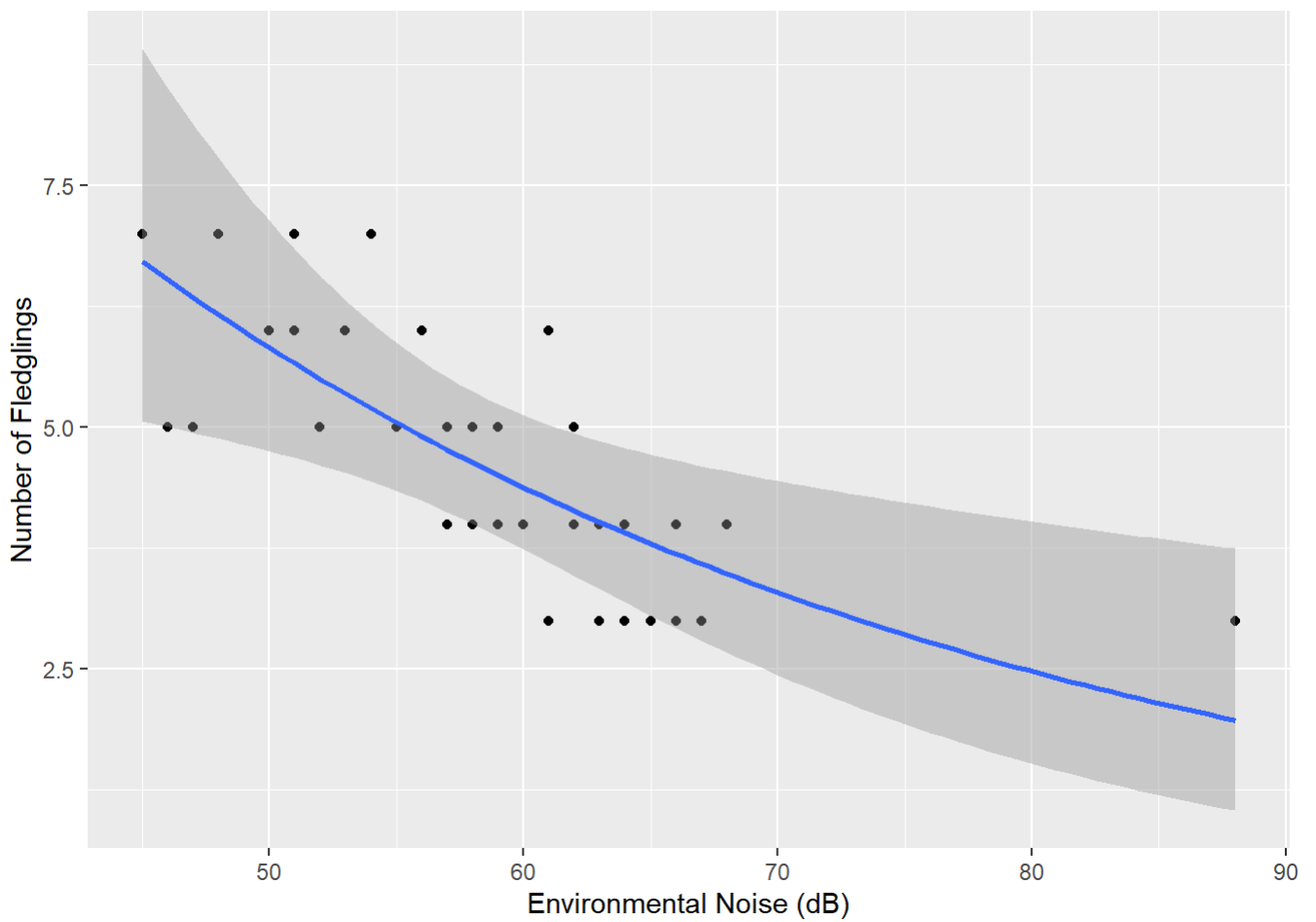


Figure 2: Number of fledglings plotted against environmental noise in dB.

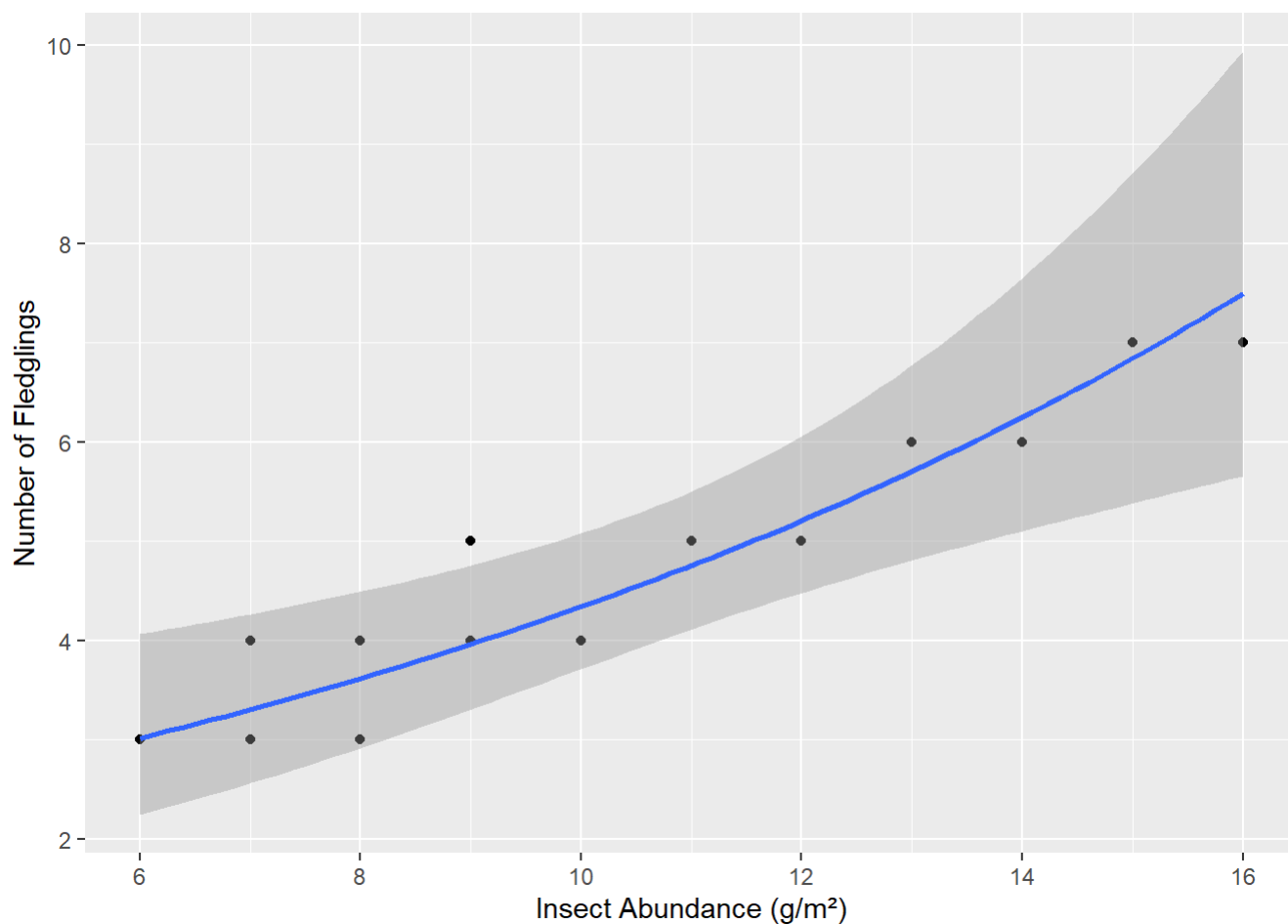


Figure 3: Number of fledglings plotted against insect abundance in g/m^2 .

3 fit candidate models & select preferred one

```
AIC(mod.00,mod.01,mod.02,mod.03,mod.04)
```

	df	AIC
mod.00	1	147.5764
mod.01	2	141.5012
mod.02	2	136.8201
mod.03	3	138.7344
mod.04	4	140.7263

```
# select mod.03 & mod.04 based on AIC
```

```
anova(mod.03, mod.04, test = "Chisq")
```

Analysis of Deviance Table

```
Model 1: number_of_fledglings ~ environmental_noise_dB + Insect.Abandance_g.m2
Model 2: number_of_fledglings ~ environmental_noise_dB * Insect.Abandance_g.m2
Resid. Df Resid. Dev Df Deviance Pr(>Chi)
```

1	36	0.99276		
2	35	0.98472	1 0.0080324	0.9286

```
# the P-value of the Chisq test is > 0.05 -> interaction term not significant
```

4 validate preferred model

```
summary(mod.03)
```

5 interpret and report findings

Each 1 dB increase in noise corresponds to 0.4 % fewer fledglings (estimate = -0.004 , p = 0.771). Each 1 g/m² increase in insect biomass is associated with 8.6 % more fledglings (estimate = 0.083 , p = 0.028).

6 store session information
